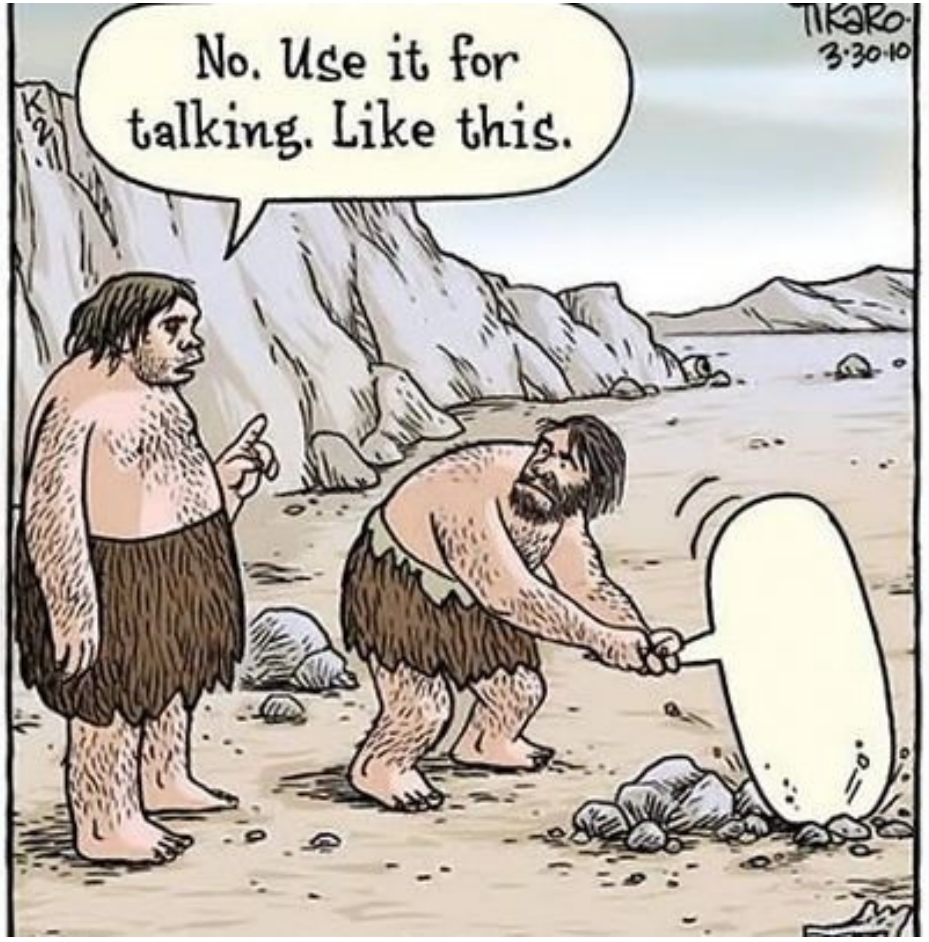
Natural Language Processing

Semantics



NLP: Semantics

“the fish is ready to eat”





Distributional semantics

Evert's slides, from 1 to 13 included

Distributional Semantic Models

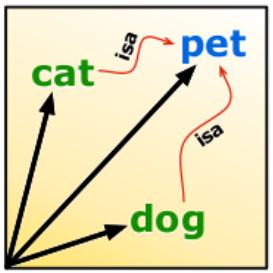
Part 1: Introduction

Stefan Evert[1]
with Alessandro Lenci[2], Marco Baroni[3] and Gabriella Lapesa[4]

[1]Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany
[2]University of Pisa, Italy
[3]University of Trento, Italy
[4]University of Stuttgart, Germany

http://wordspace.collocations.de/doku.php/course:start

Copyright © 2009–2016 Evert, Lenci, Baroni & Lapesa | Licensed under CC-by-sa version 3.0



Meaning & distribution

- "Die Bedeutung eines Wortes liegt in seinem Gebrauch."
  The meaning of a word is its use — Ludwig Wittgenstein

- "You shall know a word by the company it keeps!"
  — J. R. Firth (1957)

- Distributional hypothesis: difference of meaning correlates with difference of distribution (Zellig Harris 1954)

- "What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it [...] in everyday discourse." (Miller 1986)

Distributional semantics

Tools for Distributional Semantics: see the links in the AulaWeb Module (word2vect implementations in Python and C, with examples of use)

Frame & model-theoretic semantics

Liang's slides, from 36 to 56 included

NLU: the foundations

Natural Language Understanding: Foundations and State-of-the-Art

Percy Liang



"Foundations of ML" Bootcamp

January 26, 2017

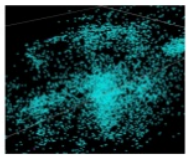
https://simons.berkeley.edu/talks/percy-liang-01-27-2017-1

NLU: the foundations

Outline



Properties of language



Distributional semantics



Frame semantics



Model-theoretic semantics



Interactive learning



Reflections

Frame & Model Theoretic Semantics

- **Tools for Frame Semantics**: see the links in the AulaWeb Module (https://framenet.icsi.berkeley.edu/fndrupal/

  examples at: https://framenet2.icsi.berkeley.edu/fnReports/data/frameIndex.xml?frame=Commerce_buy)
- **Tools for Model Theoretic Semantics**: ...not found...

Lexical semantics

Linguistic theories generally regard human languages as consisting of two parts: a lexicon, essentially a catalogue of a language's words (its wordstock); and a grammar, a system of rules which allow for the combination of those words into meaningful sentences. The lexicon is also thought to include bound morphemes, which cannot stand alone as words (such as most affixes). In some analyses, compound words and certain classes of idiomatic expressions and other collocations are also considered to be part of the lexicon.

Dictionaries represent attempts at listing, in alphabetical order, the lexicon of a given language; usually, however, bound morphemes are not included.

Lexical semantics

The units of analysis in lexical semantics are lexical units which include not only words but also sub-words or sub-units such as affixes and even compound words and phrases. Lexical units make up the lexicon.

Lexical semantics looks at how the meaning of the lexical units correlates with the structure of the language or syntax.

Lexical semantics

Lexical relations: how meanings relate to each other

Lexical items contain information about category (lexical and syntactic), form and meaning. The semantics related to these categories then relate to each lexical item in the lexicon. Lexical items can also be semantically classified based on whether their meanings are derived from single lexical units or from their surrounding environment.

Lexical items participate in regular patterns of association with each other. Some relations between lexical items include hyponymy, hypernymy, synonymy and antonymy, as well as homonymy.

Lexical relations: synonymy

- similarity of meaning
  - Leibniz: two expressions are synonymous if the substitution of one for the other never changes the truth value of a sentence in which the substitution is made
- such global synonymy is rare (it would be redundant)
  - synonymy *relative to a context*: two expressions are synonymous in a linguistic context C if the substitution of one for the other in C does not alter the truth value
  - consequence of this synonymy in terms of substitutability: words in different syntactic categories cannot be synonyms

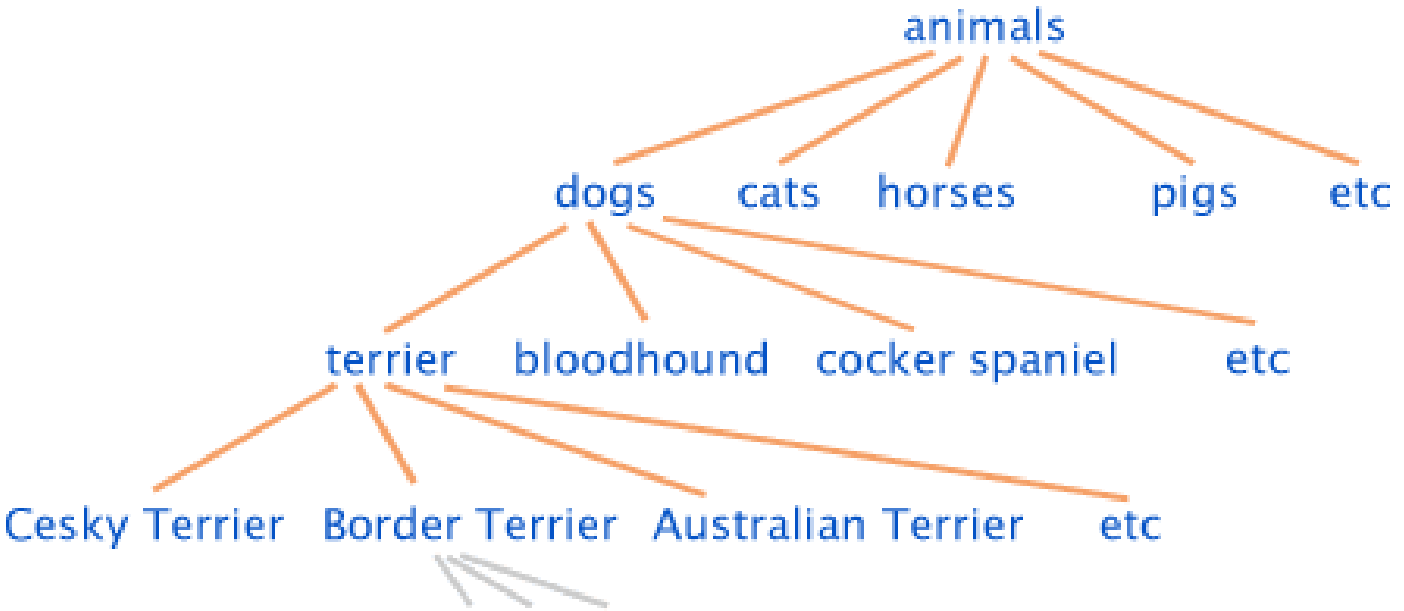
Lexical relations: antonymy

- antonym of a word x is sometimes not-x, but not always
  - *rich* and *poor* are antonyms
  - but: ***not*** *rich* does not imply *poor*
  - (because many people consider themselves neither rich nor poor)
- antonymy is a lexical relation between word forms, not a semantic relation between concepts
  Example: [rise/fall] and [ascend/descend] are pairs of antonyms

Lexical relations: hyponymy

- hyponymy is a semantic relation between word *meanings*
  - {maple} is a hyponym of {tree}
- inverse: hypernymy
  - {tree} is a hypernym of {maple}
- also called: subordination/superordination; subset/superset; IS-A relation
- test for hyponomy:
  - native speaker must accept sentences built from the frame “An x is a (kind of) y”
- called troponomy when applied to verbs

Lexical relations: hyponymy



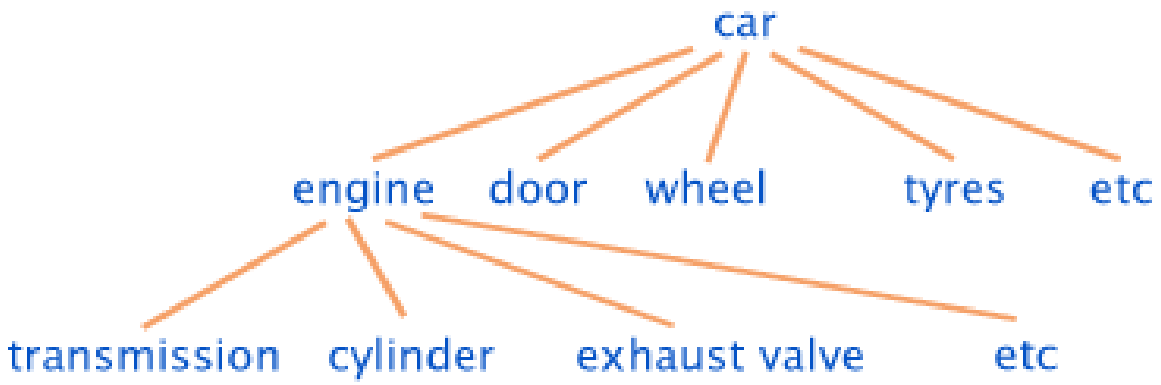
Lexical relations: meronymy

- A concept C1 is a **meronym** of a concept C2 in language L if native speakers of L accept sentences constructed from such frames as “A C2 has a C1 (as a part)”, “A C1 is a part of C2”.
- inverse relation: holonymy
- HAS-AS-PART
  - part hierarchy
  - part-of is asymmetric and (with caution) transitive

Lexical relations: meronymy

- failures of transitivity caused by different part-whole relations, e.g.
  - *A musician has an arm*.
  - *An orchestra has a musician*.
  - but: ? *An orchestra has an arm*.

Lexical relations: meronymy



Lexical relations: homonymy

HOMONYMS	HOMOPHONES	HOMOGRAPHS
multiple meaning words	*words that sound alike*	*same spelling, different pronunciation, different meanings*
the **spruce** tree... to **spruce** up...	**addition** for math **edition** of a book	**desert** = abandon **desert** = area of land
suit yourself... wore a **suit**...	I want **to** go I like it **too** One plus one is **two**	**bass** = fish **bass** = instrument
weigh on the **scale**... **scale** the wall...	**capitol** building state **capital**	**close** = nearby **close** = to shut
the price is **fair**... go to the **fair**...	pick a **flower** bake with **flour**	**bow** = to bend down **bow** = ribbon

Structured lexicons & Thesauri

- Alternative to alphabetical dictionary
- List of words grouped according to meaning
- Hierarchical organization is important
- Hierarchies familiar as taxonomies, eg in natural sciences
  - Children are “types of” and share certain properties, inherited from the father
- Similar idea for ordinary words: hyponymy and synonymy

Structured lexicons & Thesauri

- A way to show the structure of (lexical) knowledge
- Much used for technical terminology
- Can be enriched by having other lexical relations:
  - Antonyms (as well as synonyms)
  - Different hyponymy relations, not just is-a-type-of, but has-as-part/member
- Thesaurus can be explored in any direction
  - across, up, down
  - Some obvious distance metrics can be used to measure similarity between words

WordNet: History

- 1985: a group of psychologists and linguists start to develop a “lexical database”
  - Princeton University
  - theoretical basis: results from psycholinguistics and psycholexicology
  - What are properties of the “mental lexicon”?

WordNet: Global organization

- nouns: organized as topical hierarchies
- verbs: entailment relations

WordNet: Lexical semantics

- How are word meanings represented in WordNet?
  - synsets (synonym sets) as basic units
  - a word ‘meaning’ is represented by simply listing the word forms that can be used to express it
- example: senses of *board*
  - a piece of lumber vs. a group of people assembled for some purpose
  - synsets as unambiguous designators:
  - {board, plank, ...} vs. {board, committee, ...}
- Members of synsets are rarely true synonyms
  - WordNet does not attempt to capture subtle distinctions among members of the synset
  - may be due to specific details, or simply connotation, collocation

WordNet: Synsets

- synsets often sufficient for differential purposes
  - if an appropriate synonym is not available a short gloss may be used
  - e.g. {board, (a person’s meals, provided regularly for money)}
  - Preferable for cardinality of synset to be >1
  - WordNet also gives a gloss for each word meaning, and (often) an example

WordNet Search - 3.1

- WordNet home page - Glossary - Help

Word to search for: board [Search WordNet]

Display Options: (Select option to change) [Change]

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (gloss) "an example sentence"

Noun

- S: (n) **board** (a committee having supervisory powers) "*the board has seven members*"
  - *direct hyponym* / *full hyponym*
  - ***member meronym***
    - S: (n) director (member of a board of directors)
    - S: (n) board member (a member of a governing board)
  - *direct hypernym* / *inherited hypernym* / *sister term*
    - S: (n) committee, commission (a special group delegated to consider some matter) "*a committee is a group that keeps minutes and loses hours*" - *Milton Berle*
- S: (n) **board**, plank (a stout length of sawn timber; made in a wide variety of sizes and used for many purposes)
- S: (n) **board** (a flat piece of material designed for a special purpose) "*he nailed boards across the windows*"
- S: (n) **board**, table (food or meals in general) "*she sets a fine table*"; "*room and board*"
- S: (n) display panel, display board, **board** (a vertical surface on which information can be displayed to public view)
- S: (n) dining table, **board** (a table at which meals are served) "*he helped her clear the dining table*"; "*a feast was spread upon the board*"
- S: (n) control panel, instrument panel, control board, **board**, panel (electrical device consisting of a flat insulated surface that contains switches and dials and meters for controlling other electrical devices) "*he checked the instrument panel*"; "*suddenly the board lit up like a Christmas tree*"

WordNet: dimensions

DESCRIPTION

Number of words, synsets, and senses

POS	Unique Strings	Synsets	Total Word-Sense Pairs
Noun	117798	82115	146312
Verb	11529	13767	25047
Adjective	21479	18156	30002
Adverb	4481	3621	5580
Totals	155287	117659	206941

Polysemy information

POS	Monosemous Words and Senses	Polysemous Words	Polysemous Senses
Noun	101863	15935	44449
Verb	6277	5252	18770
Adjective	16503	4976	14399
Adverb	3748	733	1832
Totals	128391	26896	79450

WordNet: Lexical relations

- Nouns
  - Synonym ~ antonym (opposite of)
  - Hypernyms (is a kind of) ~ hyponym (for example)
  - Coordinate (sister) terms: share the same hypernym
  - Holonym (is part of) ~ meronym (has as part)
- Verbs
  - Synonym ~ antonym
  - Hypernym ~ troponym (eg lisp – talk)
  - Entailment (eg snore – sleep)
  - Coordinate (sister) terms: share the same hypernym
- Adjectives/Adverbs in addition to above
  - Related nouns
  - Verb participles
  - Derivational information

Word sense disambiguation

Word-sense disambiguation (WSD) is an open problem of natural language processing and ontology. WSD is identifying which sense of a word (i.e. meaning) is used in a sentence, when the word has multiple meanings.

Word sense disambiguation

To give a hint how all this works, let us consider the senses of the word “bass”, and the two sentences

- I went fishing for some sea bass.
- The bass line of the song is too weak.

Word sense disambiguation

WordNet Search - 3.1
- WordNet home page - Glossary - Help

Word to search for: bass [Search WordNet]

Display Options: (Select option to change) [Change]

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (gloss) "an example sentence"

Noun

- S: (n) **bass** (the lowest part of the musical range)
- S: (n) **bass**, bass part (the lowest part in polyphonic music)
- S: (n) **bass**, basso (an adult male singer with the lowest voice)
- S: (n) sea bass, **bass** (the lean flesh of a saltwater fish of the family Serranidae)
- S: (n) freshwater bass, **bass** (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
- S: (n) **bass**, bass voice, basso (the lowest adult male singing voice)
- S: (n) **bass** (the member with the lowest range of a family of musical instruments)
- S: (n) **bass** (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Adjective

- S: (adj) **bass**, deep (having or denoting a low vocal or instrumental range) "*a deep voice*"; "*a bass voice is lower than a baritone voice*"; "*a bass clarinet*"

Word sense disambiguation

To a human, it is obvious that the first sentence is using the word "bass (fish)" and the second sentence, the word "bass (instrument)" is being used. Developing algorithms to replicate this human ability can often be a difficult task, as is further exemplified by the implicit equivocation between "bass (sound)" and "bass (musical instrument)".

WSD: Lesk Algorithm

The Lesk algorithm is a classical algorithm for word sense disambiguation introduced by Michael E. Lesk in 1986 (“ Automatic sense disambiguation using machine readable dictionaries: how to tell a pine cone from an ice cream cone”. In Proc. of SIGDOC '86, ACM).

The Lesk algorithm is based on the assumption that words in a given "neighborhood" (section of text) will tend to share a common topic. A simplified version of the Lesk algorithm is to compare the dictionary definition of an ambiguous word with the terms contained in its neighborhood.

WSD: Lesk Algorithm

An implementation might look like this:

- for every sense of the word being disambiguated one should count the amount of words that are in both neighborhood of that word and in the dictionary definition of that sense
- the sense that is to be chosen is the sense which has the biggest number of this count

WSD: Lesk Algorithm

A frequently used example illustrating this algorithm is for the context "pine cone". The following dictionary definitions are used:

```
PINE
1. kinds of evergreen tree with needle-shaped leaves
2. waste away through sorrow or illness
```

```
CONE
1. solid body which narrows to a point
2. something of this shape whether solid or hollow
3. fruit of certain evergreen trees
```

As can be seen, the best intersection is Pine #1 ⋂ Cone #3 = 2.

WSD: Lesk Algorithm

Extension of the Lesk algorithm for working with WordNet: Satanjeev Banerjee and Ted Pedersen. “An Adapted Lesk Algorithm for Word Sense Disambiguation Using WordNet”, Lecture Notes In Computer Science; Vol. 2276, Pages: 136 - 145, 2002. ISBN 3-540-43219-1

WSD: Lesk Algorithm

Lesk's approach is very sensitive to the exact wording of definitions, so the absence of a certain word can radically change the results.

Further, the algorithm determines overlaps only among the glosses of the senses being considered. This is a significant limitation in that dictionary glosses tend to be fairly short and do not provide sufficient vocabulary to relate fine-grained sense distinctions.

Recently, a lot of works appeared which offer different modifications of this algorithm. These works use other resources for analysis (thesauruses, synonyms dictionaries or morphological and syntactic models).

Named Entity Recognition

Named-entity recognition (NER) seeks to locate and classify named entities in text into pre-defined categories such as the names of persons, organizations, locations, expressions of times, quantities, monetary values, percentages, etc.

Most research on NER systems has been structured as taking an unannotated block of text, such as this one:

Jim bought 300 shares of Acme Corp. in 2006.

And producing an annotated block of text that highlights the names of entities:

[Jim]Person bought 300 shares of [Acme Corp.]Organization in [2006]Time.

Named Entity Recognition

Online:

- http://nlp.stanford.edu:8080/ner/process
- http://textanalysisonline.com/spacy-named-entity-recognition-ner

For Italian (not only NER!):

- http://www.italianlp.it/
- http://parli.di.unito.it/link_it.html

Named Entity Recognition

NER systems have been created that use linguistic grammar-based techniques as well as statistical models, i.e. machine learning. Hand-crafted grammar-based systems typically obtain better precision, but at the cost of lower recall and months of work by experienced computational linguists.

Statistical NER systems typically require a large amount of manually annotated training data.

Semisupervised approaches have been suggested to avoid part of the annotation effort.

Named Entity Recognition

NER systems have been created that use linguistic grammar-based techniques as well as statistical models, i.e. machine learning. Hand-crafted grammar-based systems typically obtain better precision, but at the cost of lower recall and months of work by experienced computational linguists.

Statistical NER systems typically require a large amount of manually annotated training data.

Semisupervised approaches have been suggested to avoid part of the annotation effort.

Terminology



In the field of information retrieval, **precision** is the fraction of retrieved documents that are relevant to the query:

$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

For example, for a text search on a set of documents, precision is the number of correct results that have been obtained, divided by the number of all returned results.

Terminology



In information retrieval, **recall** is the fraction of the relevant documents that are successfully retrieved.

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

For example, for a text search on a set of documents, recall is the number of correct results that have been obtained divided by the number of results that should have been returned.

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Example: a query for documents about topic T inside a corpus C is made and the result known to be correct is:
D1, D2, D4, D5, D6, D7, D8 all deal with T, and no other document in C deals with T.

Algorithm A1 returns documents D1, D2, D3, D4.

precision(A1) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1, D2, D3, D4} |/
| {D1, D2, D3, D4} |
= 3 / 4 = 75%

recall(A1) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1, D2, D3, D4} |/
| {D1, D2, D4, D5, D6, D7, D8} |
= 3 / 7 = 43%

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Example: a query for documents about topic T inside a corpus C is made and the result known to be correct is:
D1, D2, D4, D5, D6, D7, D8 all deal with T, and no other document in C deals with T.

Algorithm A2 returns documents D1, D2, D3, D4, D5, D6, D7, D8.

precision(A1) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1, D2, D3, D4, D5, D6, D7, D8} |/
| {D1, D2, D3, D4, D5, D6, D7, D8} |
= 7 / 8 = 87.5%

recall(A1) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1, D2, D3, D4, D5, D6, D7, D8} |/
| {D1, D2, D4, D5, D6, D7, D8} |
= 7 / 7 = 100%

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Example: a query for documents about topic T inside a corpus C is made and the result known to be correct is:
D1, D2, D4, D5, D6, D7, D8 all deal with T, and no other document in C deals with T.

Algorithm A3 returns document D1 only

precision(A3) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1} |/
| {D1} |
= 1 / 1 = 100%
recall(A3) = | {D1, D2, D4, D5, D6, D7, D8} ∩ {D1} |/
| {D1, D2, D4, D5, D6, D7, D8} |
= 1 / 7 = 14%

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Example: a query for documents about topic T inside a corpus C is made and the result known to be correct is:
D1, D2, D4, D5, D6, D7, D8 all deal with T, and no other document in C deals with T.

Algorithm A4 returns all and only the relevant documents

precision(A4) = | {relevant} ∩ {relevant} |/
| {relevant} |
= 7 / 7 = 100%
recall(A4) = | {relevant} ∩ {relevant} |/
| {relevant} |
= 7 / 7 = 100%

Terminology



$$\text{precision} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{retrieved documents}\}|}$$

$$\text{recall} = \frac{|\{\text{relevant documents}\} \cap \{\text{retrieved documents}\}|}{|\{\text{relevant documents}\}|}$$

Example: a query for documents about topic T inside a corpus C is made and the result known to be correct is:
D1, D2, D4, D5, D6, D7, D8 all deal with T, and no other document in C deals with T.

Algorithm A5 returns 1000 documents, including the 7 relevant ones

precision(A5) = | {relevant} ∩ {retrieved} |/
| {retrieved} |
= 7 / 1000 = 0.7%
recall(A5) = | {relevant} ∩ {retrieved} |/
| {relevant} |
= 7 / 7 = 100%